



Southeast University

Department of Computer Science and Engineering

LAB Report

Course Name: Computer networking LAB

Course Code: CSE 343.2

Experiment No.: 03

Experiment Name: HTTP, DNS and DHCP Configuration

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❑ Experiment no: 03

❑ Experiment Name: Basic LAN Configuration with Router's Interface using HTTP, DNS Protocols with DHCP Configuration.

❑ Objective: The objective of this experiment is to connect two distinct networks using a router and configure it to automatically assign IP addresses to end-devices via a DHCP server. Moreover the experiment aims to deploy web (HTTP) and DNS servers to host and resolve a local website. Ultimately verifying successful inter-network communication and webpage accessibility across the simulated topology.

❑ Network Setup: The topology has two different networks connected with Router. There is a HTTP web server and DNS server too.

Device / Interface	Network / IP Addresses	Subnet mask	Gateway
Gig 0/0	192.168.1.0	255.255.255.0	192.168.1.1
Gig 0/1	192.60.1.0	255.255.255.0	192.60.1.1
DNS server	192.168.1.2	255.255.255.0	192.168.1.1
HTTP server	192.168.1.3	255.255.255.0	192.168.1.1
End-devices gig 0/0	DHCP	255.255.255.0	192.168.1.1
End-devices gig 0/1	DHCP	255.255.255.0	192.60.1.1

4. CLI Commands on Router's Interface: Open Desktop tab on Laptop and then terminal interface to give commands.

- Interface gig 0/0:

```
Router> Enable
```

```
Router# Configure terminal
```

```
Router(config)# Interface gig 0/0
```

```
Router(config-if)# IP address 192.168.1.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

- Interface gig 0/1:

```
Router(config)# Interface gig 0/1
```

```
Router(config-if)# IP address 192.60.1.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

4. DHCP Configuration: for Interface gig 0/0

```
Router(config)# IP dhcp Pool dhcp-one
```

```
Router(dhcp-config)# Network 192.168.1.0 255.255.255.0
```

```
Router(dhcp-config)# default-router 192.168.1.1
```

```
Router(dhcp-config)# DNS-server 192.168.1.2
```

```
Router(dhcp-config)# IP dhcp excluded-address  
192.168.1.1 192.168.1.3
```

```
Router(dhcp-config)# exit
```

- DHCP for interface gig 0/1:

Router(config)# ip dhcp pool dhcp-two

Router(dhcp-config)# network 192.168.1.0 255.255.255.0

Router(dhcp-config)# default-router 192.168.1.1

Router(dhcp-config)# dns-server 192.168.1.2

Router(dhcp-config)# exit

Router(config)# exit

Router# write memory

☐ Server Setup:

- Web Server: Go to desktop > IP configuration

- IP → 192.168.1.3

- Gateway → 192.168.1.1

- DNS server → 192.168.1.2

Then go to services > HTTP and turn it on.

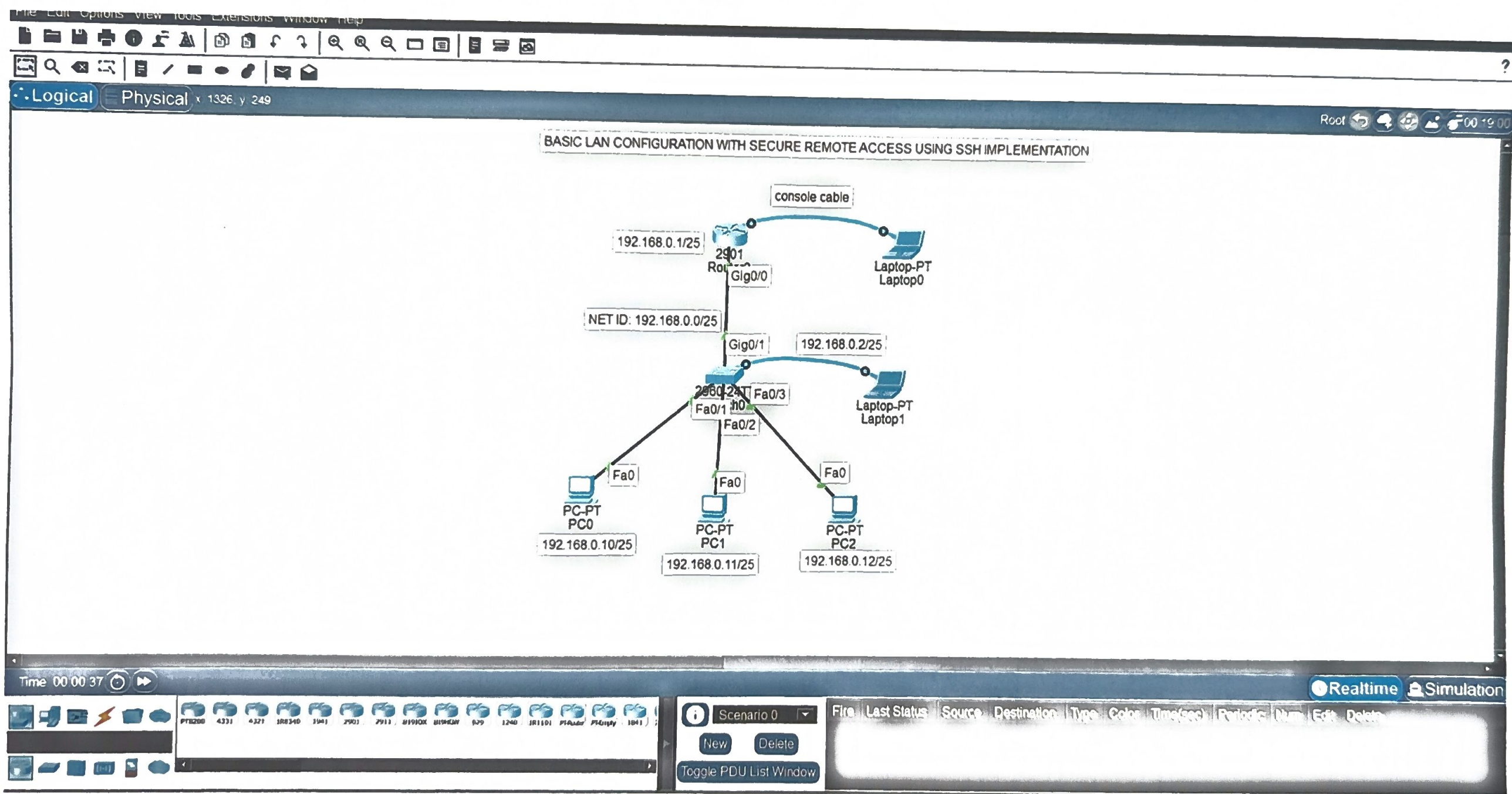
- DNS server: Go to desktop > IP configuration

- IP → 192.168.1.2

- Gateway → 192.168.1.1

- DNS server → 192.168.1.2

Then go to services > DNS, turn it on and add a new record sev.edu.bd, address 192.168.1.3.



- End-devices: Go to desktop > IP Configuration. Change the setting from static to DHCP. They will get an IP automatically.

Testing the network:

- Check DHCP: Open any end-device and saw it successfully received an IP address like (192.168.1.4) automatically.
- Website Test: Open any end-device and go to web browser and type sev.edu.bd. The web page loaded successfully, meaning the router is passing traffic correctly.

Conclusion: In this experiment we successfully connected two different networks using a Cisco Router. We learned how to configure DHCP. We also set up web and DNS servers and proved that everything was communicating properly by pinging across the networks and opening a webpage. This experiment helped us understand how real-world basic LANs and services operate.